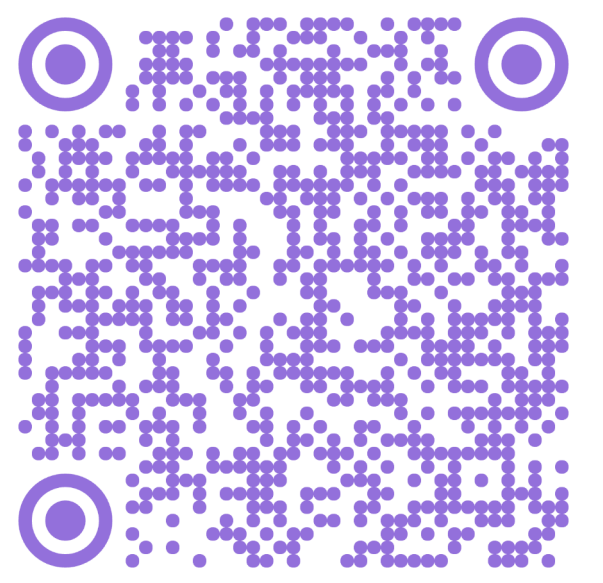


TractorBeam: Personalized AI Sensemaking Support via Collaborative Machine Annotation

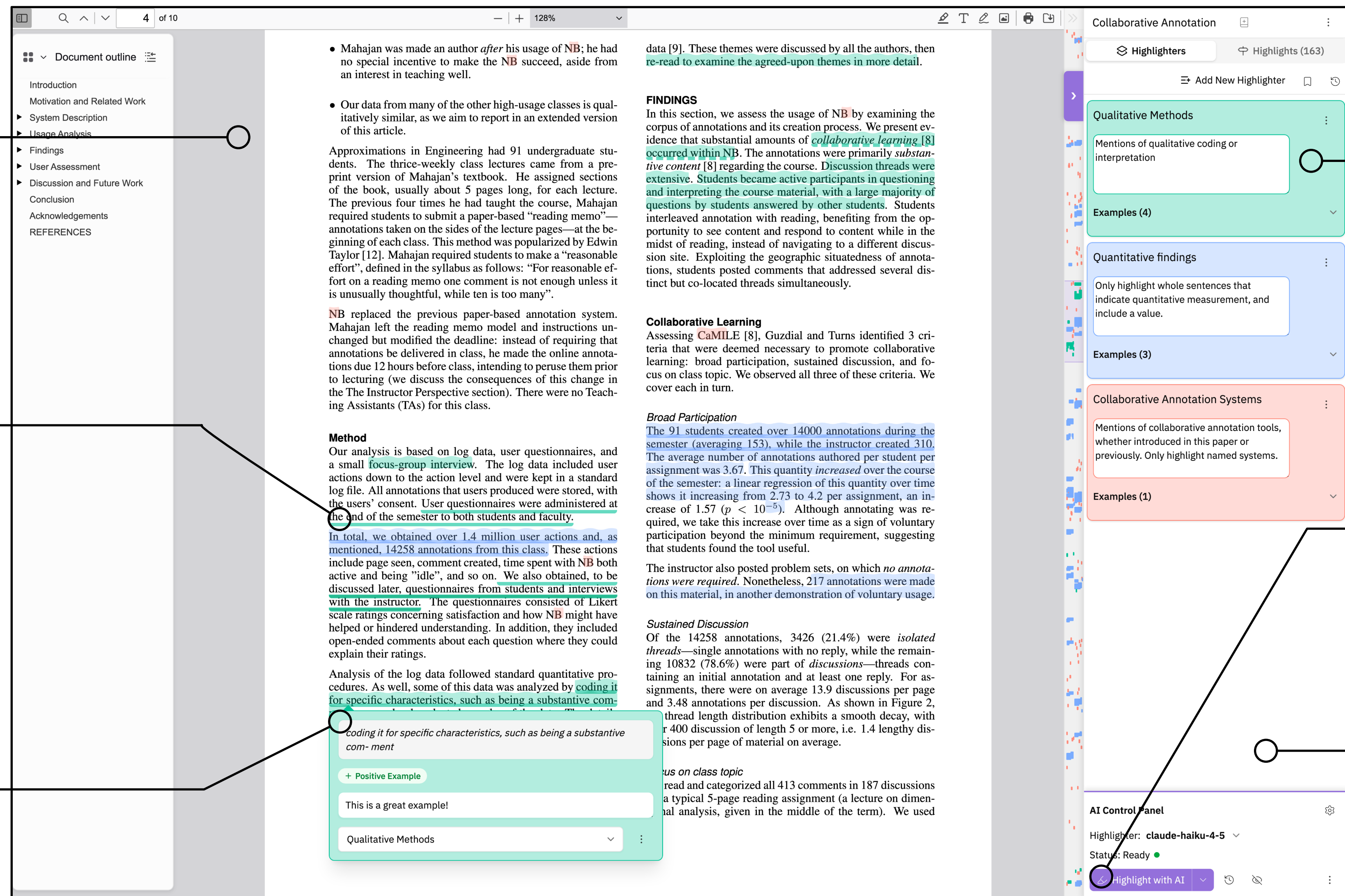


Sireesh Gururaja, Jordan Taylor, and Emma Strubell

TractorBeam instruments Mozilla's pdf.js viewer, letting sensemaking start where it usually does: **reading**.

Highlight style indicates provenance and certainty: underlines indicate user annotations, squiggly indicates machine highlights

User and machine highlights are simultaneously displayed on the PDF, allowing users to **evaluate** and **offer feedback** to improve machine highlights



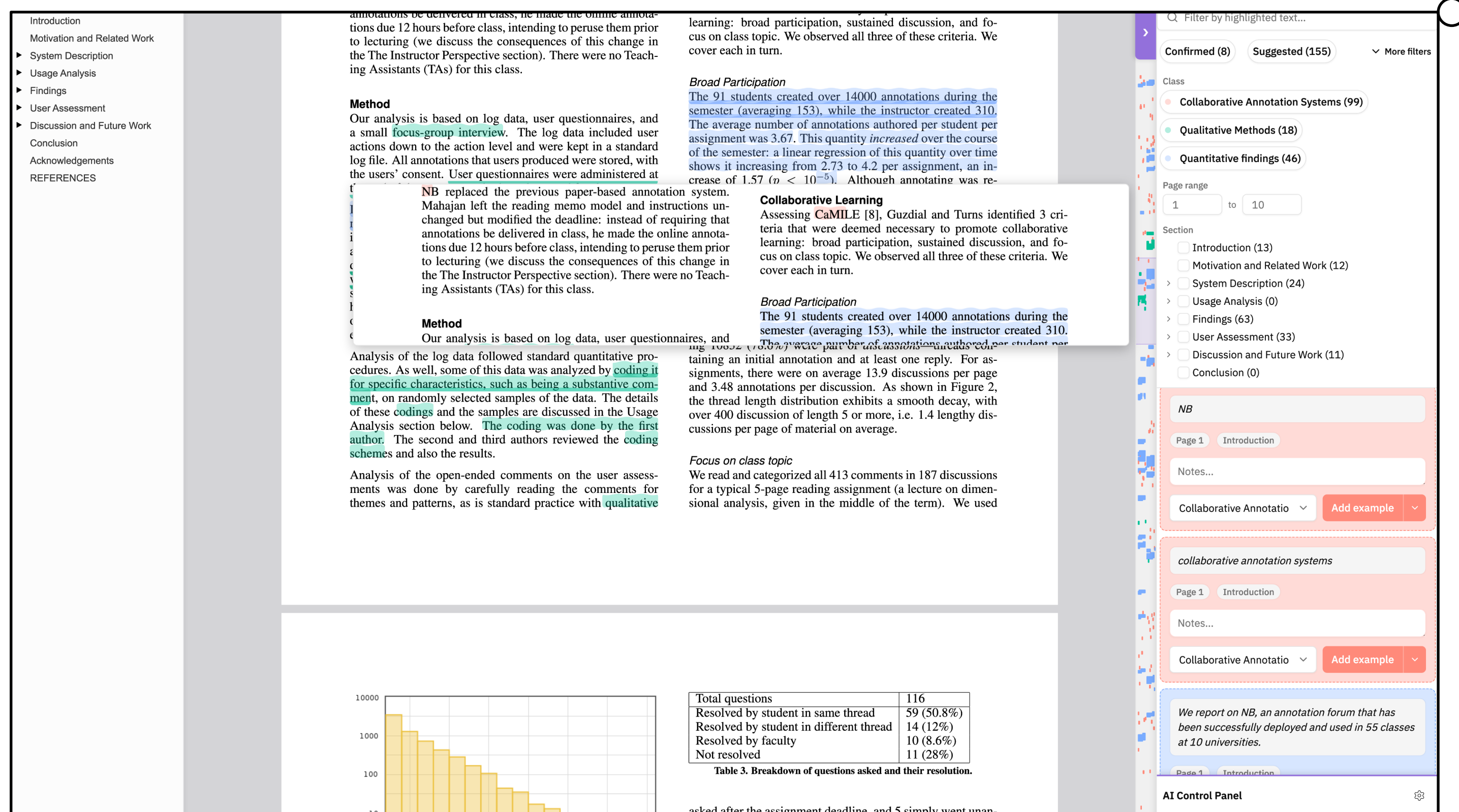
Users specify what they want to look for in **highlighters**. Highlighters are versioned, and editable anytime, to facilitate users **evolving their conceptual model**.

AI Highlighting is always initiated by the user, and is based on the users **highlighters**.

Users can configure any OpenAI- or Anthropic-compliant model, or even run a model in the browser.

Highlight: Navigation

User Study



6 Users across materials science (2), materials informatics (1), environmental policy scholarship (2) and broadcast journalism (1)

Users felt they could **iterate**

"I would say it's the second pass that I saw functionality on" (P3)

"Once we described it, it is good" (P2)

"just needed one negative example of the first author, and it corrected that on another attempt" (P1)

Highlight: Managing Data

Highlighter	PDF	Highlighted Text	Notes	Open
Qualitative Methods	https://people.csail.mit.edu/...	collaborative learning [8] occurred within NB	—	Open
Qualitative Methods	https://people.csail.mit.edu/...	coding it for specific characteristics, such as being a substantive com- ment	This is a great example!	Open
Qualitative Methods	https://dl.acm.org/doi/pdf/...	User questionnaires were administered at the end of the semester to both students and faculty	—	Open
Qualitative Methods	https://people.csail.mit.edu/...	User questionnaires were administered at the end of the semester to both students and faculty	—	Open
Qualitative Methods	https://people.csail.mit.edu/...	We also obtained, to be discussed later, questionnaires from students and interviews with the instructor.	—	Open
Qualitative Methods	file:///Users/sireeshgururaja/...	We identify specific reading behaviors that are predictive of in-class exam performance.	—	Open
Quantitative Findings	https://people.csail.mit.edu/...	We report on NB, an annotation forum that has been successfully deployed and used in 55 classes at 10 universities. Students use NB to hold threaded discussions in the margins of online class material	—	Open
Quantitative Findings	https://people.csail.mit.edu/...	The 91 students created over 14000 annotations during the semester (averaging 153), while the instructor created 310.	—	Open
Quantitative Findings	https://dl.acm.org/doi/pdf/...	We report on NB, an annotation forum that has been successfully deployed and used in 55 classes at 10 universities. Students use NB to hold threaded discussions in the margins of online class material	—	Open
Quantitative Findings	https://dl.acm.org/doi/pdf/...	91 undergraduate stu- dents	—	Open
Quantitative Findings	https://dl.acm.org/doi/pdf/...	The 91 students created over 14000 annotations during the semester (averaging 153), while the instructor created 310.	—	Open
Quantitative Findings	https://people.csail.mit.edu/...	In total, we obtained over 1.4 million user actions and, as mentioned, 14258 annotations from this class.	—	Open
Quantitative Findings	file:///Users/sireeshgururaja/...	60-80% of students do not read the textbook before coming to class	—	Open
Quantitative Findings	file:///Users/sireeshgururaja/...	Figure 9 shows the extent to which students complete pre-class reading assignments over the two semesters that Persall was being used. In each semester, approximately 60% of students completed every one of the 17 reading assignments.	—	Open
Quantitative Findings	file:///Users/sireeshgururaja/...	Figure 9 shows that in S15, approximately 90% of students completed all but a couple of reading assignments; in F16, 90% of students completed all but a couple of reading assignments.	—	Open
Collaborative Annotation Systems	https://people.csail.mit.edu/...	NB	—	Open
Collaborative Annotation Systems	https://dl.acm.org/doi/pdf/...	NB	—	Open

Future Work

- **Multimodality**: Different processing for text and other page elements, like tables or charts, ignoring e.g. footnotes
- **User Agency**: How can we empower individual users to confidently create, evaluate, and edit model behavior?
- **Preserving interpretive work**: How can we build technology that allows for ambiguity, subjectivity, and positionality?